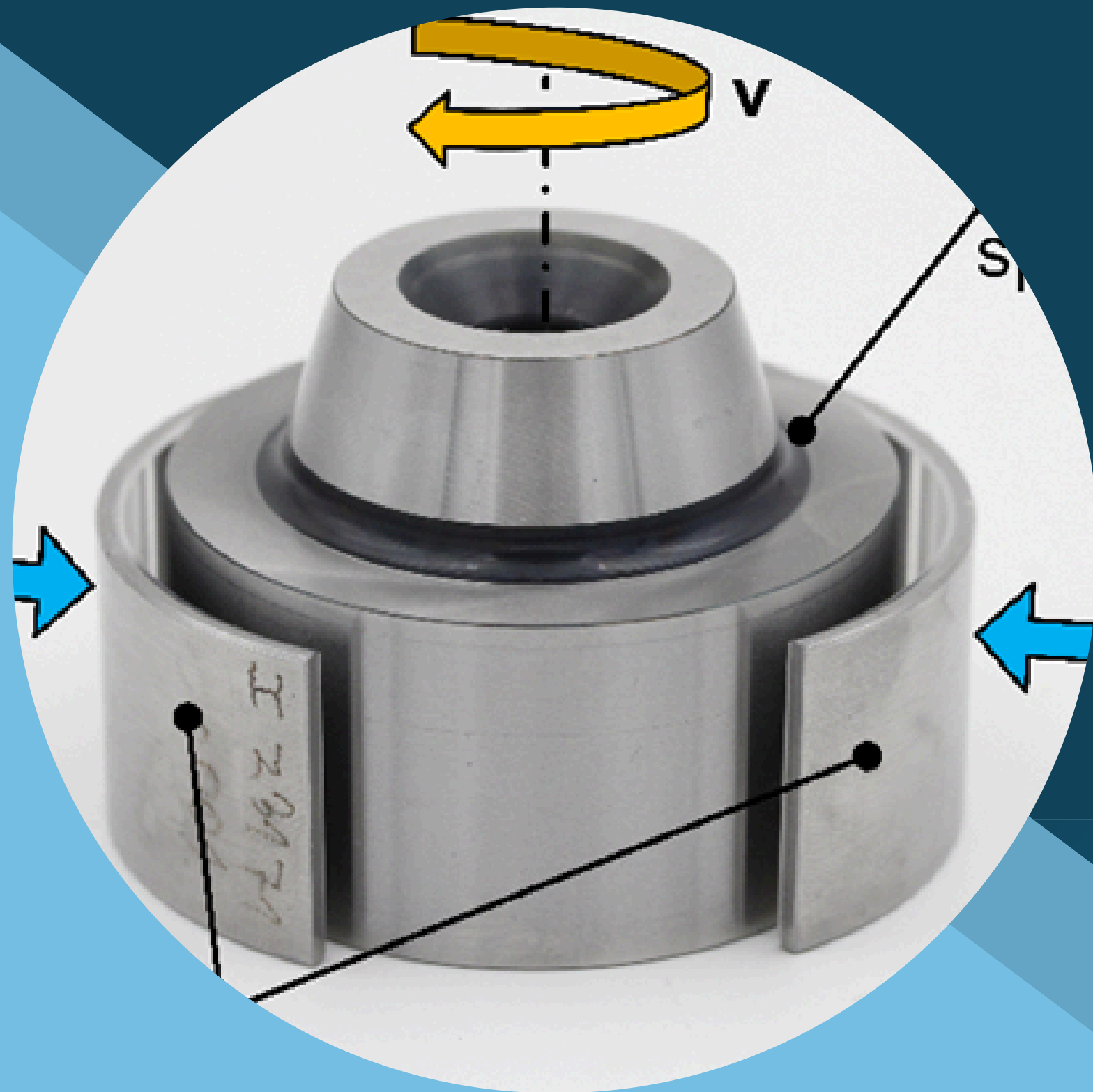




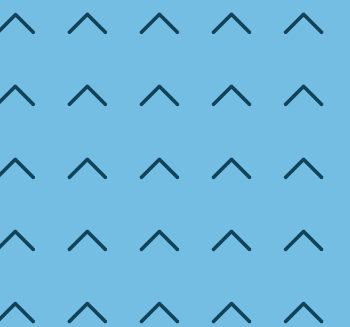
PREDICTIVE MAINTENANCE OF BEARING SHELLS

Chair of Cyber-Physical-Systems

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MOTIVATION

- 1- Journal bearing failures cause high maintenance costs and downtime.
- 2- Friction behavior is a key indicator of bearing condition, but it is difficult to model due to noisy and interacting sensor signals.
- 3- Machine learning can help capture these relationships.





RESEARCH QUESTION



Can machine learning models predict the coefficient of friction from sensor data, and which model performs better?

DATA AND PROBLEM



Dataset: Journal Bearing Adapter experiments.

Measurements: torque, load, speed, temperature, and acoustic emission.

Task: regression problem with the coefficient of friction as the target variable.

METHODS

The data was preprocessed and normalized, and physically motivated features were engineered.

Two regression models were trained and compared:

1- Random Forest

2- MLP

3- Model performance was evaluated using RMSE, MAE, and R^2 .

RESULTS

Random Forest model - very high accuracy on the test set ($R^2 \approx 0.99997$).
MLP model - poor generalization with negative R^2 .

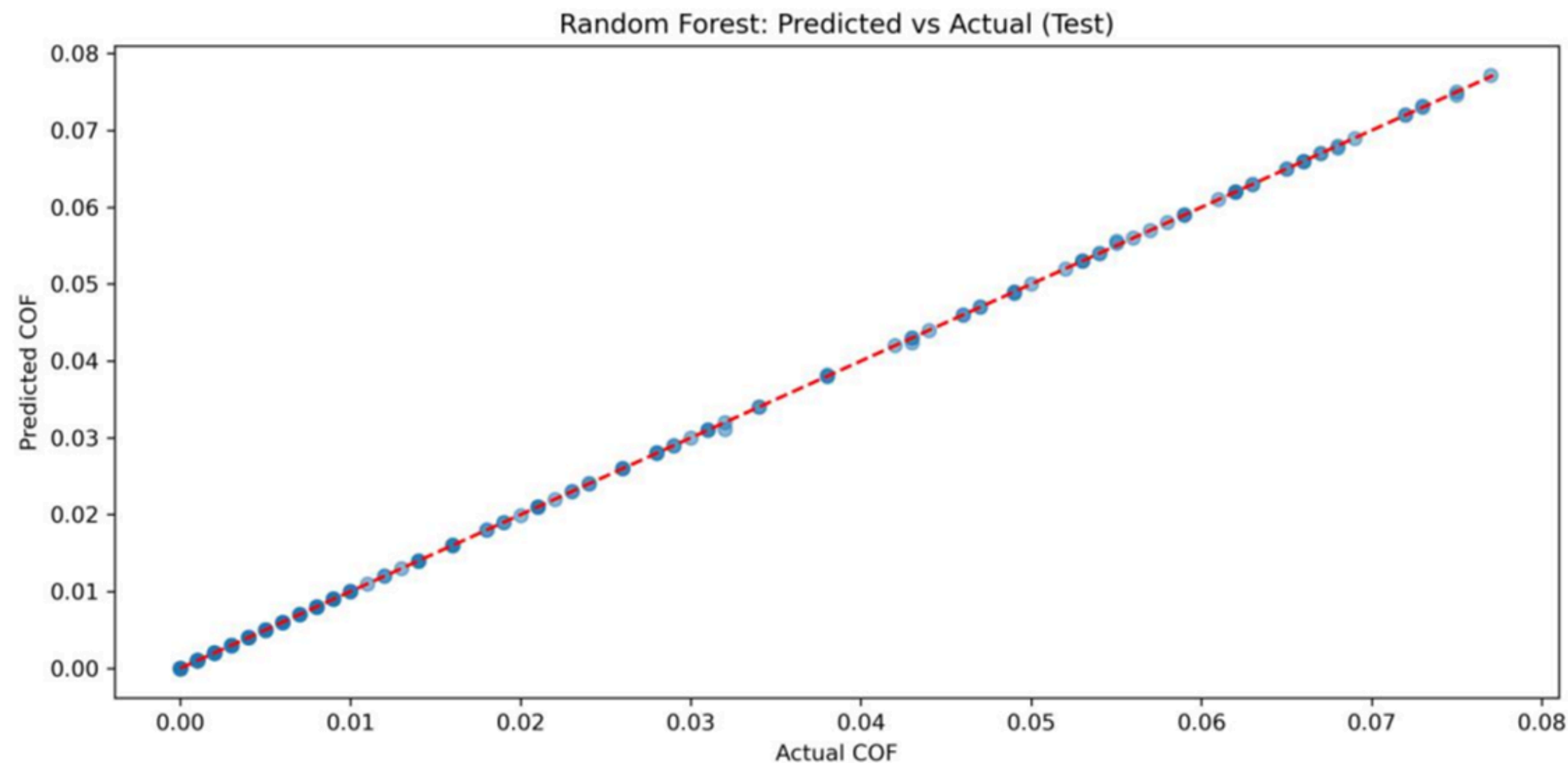
Random Forest
clearly outperforms
MLP on all metrics.

Feature importance analysis showed torque as the dominant predictor, which matches physical expectations.

Model	RMSE (Test)	MAE (Test)	R^2 (Test)
Random Forest	1.20×10^{-4}	3.17×10^{-5}	0.99997
MLP Regressor	4.82×10^{-2}	1.49×10^{-2}	-4.07

MODEL PERFORMANCE ON TEST DATASET

The predictions match the actual values, indicating very high accuracy.



RANDOM FOREST: PREDICTED VS ACTUAL COF

CONCLUSION



Random Forest is well suited for friction prediction in this dataset.
Direct bearing failure prediction was not possible due to limited wear
variability,
but accurate friction prediction is a meaningful step toward predictive
maintenance.

THANK YOU